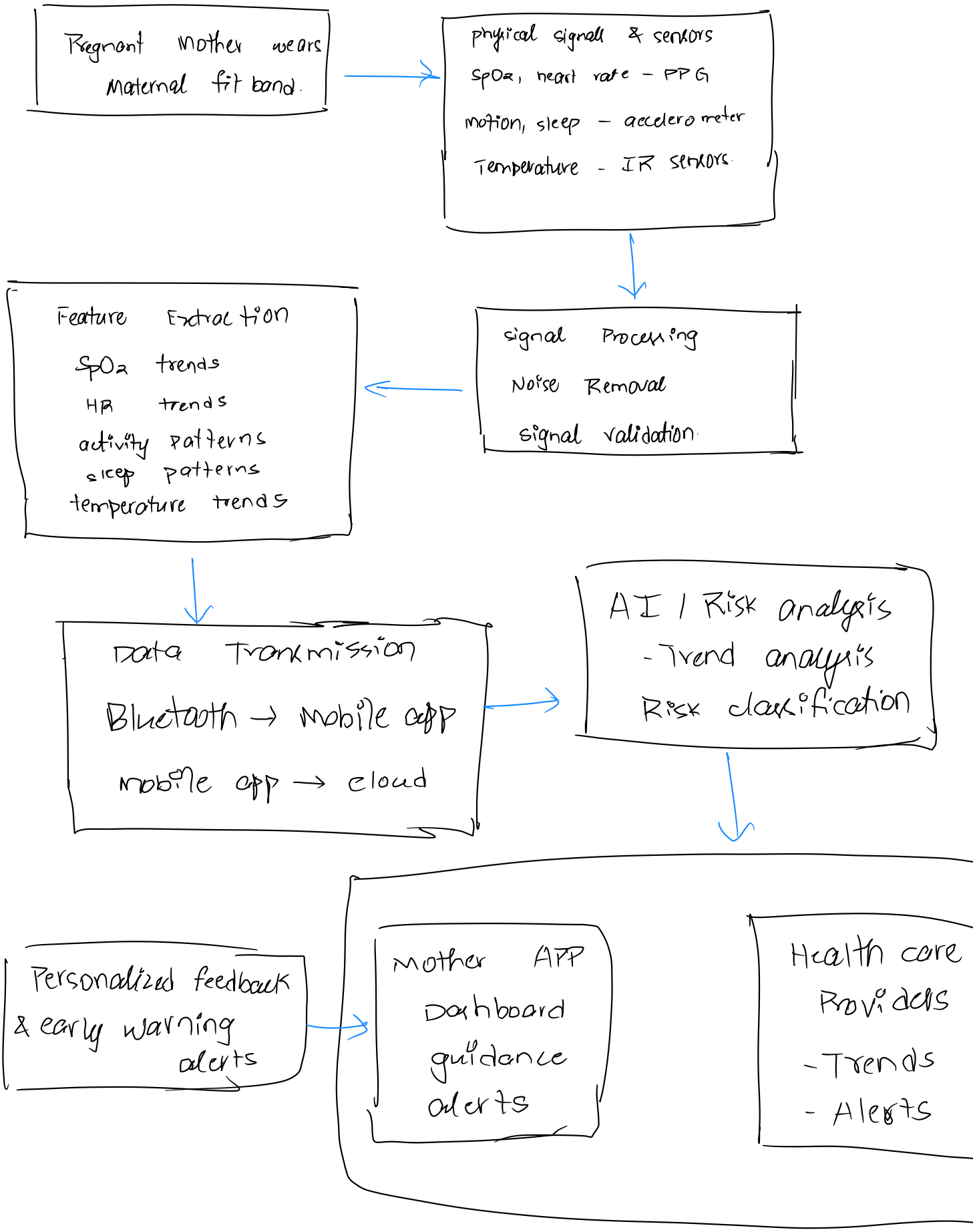

Pregnant Mother wears Maternal fit band.
physical signals & sensors
SpO2, heart rate - PPG
motion, sleep - accelerometer
Temperature - IR sensors
signal Processing
Noise Removal
signal validation
Feature Extraction
SpO2 trends
HR trends
activity patterns
sleep patterns
temperature trends
Data Transmission
Bluetooth → Mobile app
mobile app → cloud
AI / Risk analysis
- Trend analysis
Risk classification
Personalized feedback & early warning alerts
Mother APP
Dashboard
guidance
alerts
Health care Providers
- Trends
- Alerts

Hardware components:

PPG sensor — Heart Rate, SpO2

MAX 30102

3 Axis accelerometer — sleep activity

MPU 6050

ESP32 Microcontroller — wireless transfer of data